

Quiz 13

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Transform the following equation into an equivalent system of first-order differential equations:

$$x''' = x' + \cos t$$

$$x_1 = x, \quad x_2 = x', \quad x_3 = x''$$

$$\Rightarrow \begin{cases} x_1' = x_2 \\ x_2' = x_3 \\ x_3' = x_2 + \cos t \end{cases}$$

Problem 2. (6 points) Find a general solution of the system:

$$x' = y, \quad y' = -9x + 6y$$

$$x'' = y' = -9x + 6y = -9x + 6x'$$

$$\Rightarrow x'' - 6x' + 9x = 0$$

$$r^2 - 6r + 9 = 0 \Rightarrow (r-3)^2 = 0 \Rightarrow r_1 = r_2 = 3$$

$$\Rightarrow x(t) = e^{3t}(C_1 + C_2 t)$$

$$\begin{aligned} y(t) = x' &= e^{3t}(C_2) + 3e^{3t}(C_1 + C_2 t) \\ &= e^{3t}(C_2 + 3C_1 + 3C_2 t) \end{aligned}$$